

Majorana zero modes in silicon: quantified constraints on the conduction band and a conditional hole-based route

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Abstract

We assess the feasibility of Majorana zero modes (MZMs) in silicon within a Bogoliubov-de Gennes framework whose validation chain, derivations, parameter tables, and convergence studies are given in the Supplementary Material. Conduction-band wires face three independent, quantified obstructions: spin-orbit coupling (best intrinsic topological gap $\sim 1.5 \mu\text{eV}$; engineered SOC of $\alpha \gtrsim 0.02 \text{ eV \AA}$ required), a $g=2$ ceiling that, within our single-channel model, prevents induced gaps above $116 \mu\text{eV}$ from reaching the topological phase at attainable fields (quantifying the optimal-coupling rule of Cole *et al.*), and valley physics—where we prove an exact two-band equivalence for smooth interfaces (Supplement S2) and derive that each single-atomic interface step acts as a near- π Josephson junction in the band-projected pairing ($\Delta\varphi = 2k_0a/4 \approx 0.85\pi$, Supplement S3), with a numerically determined bound state that caps the topological gap at $\sim 18\%$ from the very first step. For the valence band we derive the gate-field covariation of $(\alpha, \hat{g}, m^*)$ from a four-band Luttinger-Kohn fin model: the field that builds direct-Rashba SOC (α up to $0.054 \text{ eV \AA}$, spin-orbit lengths 32–101 nm, matching FinFET measurements) suppresses the wire-axis g below 1.1 while raising g_z to 2.4–4.2. Field-orientation maps then select the candidate designs. We label each input by its evidentiary status. *Measured*: Si:B critical fields ($B_{c2} \approx 0.1\text{--}0.4 \text{ T}$), FinFET hole l_{so} and g -tensors. *Derived*: the LK operating curve and all topological-gap numbers. *Hypothetical*: Pauli-limited thin-film Si:B and a hard-gap Al/Si-hole interface. On measured inputs only, a thick Si:B parent with tilted field supports $10\text{--}19 \mu\text{eV}$. If $10\text{--}20 \text{ nm}$ Si:B films reach their Pauli limit (a fabrication target consistent with measured coherence lengths, but not yet demonstrated), $21\text{--}27 \mu\text{eV}$; if a hard-gap Al/Si-hole interface existed (none does), $34 \mu\text{eV}$ at the metallization-renormalized center point and up to $55 \mu\text{eV}$ *bare* over field orientations. The hole route is therefore a *conditional* feasibility result — a hypothesis-generating design study whose status two standard measurements would largely settle — not an established platform. Four realism controls sharpen these numbers: a dynamic tunneling self-energy lowers the center-point gaps to $7.6 / 14.9 / 24.2 \mu\text{eV}$ (measured-Si:B / Pauli-Si:B[†] / Al[†]), validating the static-renormalized caricature directionally; orbital magnetic coupling is negligible at fin widths ($< 0.05\%$); the step-junction mechanism survives pairing-channel mixing in the conservative direction; and correlated electrostatic disorder is $4\text{--}8\times$ harsher than the iid baseline at matched RMS, making gate smoothness the binding fabrication constraint. Transport calculations show our spectral disorder thresholds are conservative, identify an abrupt class-D breakdown mode, and warn that step-disordered wires become Anderson insulators whose hard-looking transport gaps mask degraded local protection. Quasiparticle poisoning restricts the measured-field all-silicon stack to detection (not parity-coherent) experiments unless the parent gap is engineered. Two measurements would do the most to settle the platform: the parallel critical field of thin Si:B films, and the wire-axis hole g -tensor in device geometry.

1 Introduction

Semiconductor–superconductor hybrid nanowires remain the most studied route to MZMs [1, 2, 3], and a decade of experiments—from the first InSb signatures [4] through the disorder-driven retractions and reappraisals [5] to protocol-based standards [6]—has made one lesson clear: material quality and honest accounting of trivial explanations dominate the problem. Silicon, the most controllable semiconductor in existence, has been conspicuously absent from this program, for reasons usually stated qualitatively: weak spin–orbit coupling (SOC), a small g -factor, and conduction-band valley degeneracy.

This paper makes those reasons quantitative, finds that one of them hides a qualitatively new failure mechanism, and shows that none of them applies to silicon *holes*. Our contributions: (i) a quantified electron verdict (Secs. 3.1, 3.2); (ii) an exact unitary equivalence making smooth-interface valley splitting benign under the physical inter-valley pairing channel, and the identification of single-atomic interface steps as near- π Josephson junctions that sharply degrade protection from the first step (single-step cap $\sim 18\%$) (Sec. 3.3); (iii) a Luttinger–Kohn (LK) derivation of the gate-field covariation of hole parameters, which erodes free-parameter optimism but preserves a viable design (Sec. 4); and (iv) orientation- resolved feasibility maps for two silicon-compatible parents—laser-doped superconducting Si:B [18, 19, 21, 22] and Al films—including thin-film design targets, metallization renormalization, and quasiparticle-poisoning limits (Sec. 4.2). Throughout, spectral diagnostics are cross-checked by recursive-Green’s-function transport (Sec. 5).

2 Model and methods

We use the standard single-channel BdG Hamiltonian

$$H = \left(\frac{\hbar^2 k^2}{2m^*} - \mu + V(x) \right) \sigma_0 + \alpha k \sigma_z + E_Z \sigma_x \quad (\text{normal part}), \quad \Delta i \sigma_y \quad (\text{pairing}), \quad (1)$$

discretized with exact particle–hole symmetry by construction, topological for $E_Z^2 > \Delta^2 + \mu^2$. Validation (repository Fig. 1): gap closing at the analytic boundary; end-localized zero modes; exponential length splitting; finite-wire vs analytic bulk-gap agreement. Valley physics uses an $8N$ basis with *inter-valley* singlet pairing $\Delta(\nu_x \otimes i\sigma_y)$ —the physical zero-momentum channel since the relevant valleys are time-reversed partners—with arbitrary complex valley-orbit (VO) profiles $\lambda(x)$ and a gauge-covariant phase-locked Dresselhaus-type SOC option. Hole parameters derive from a four-band LK model of the fin cross-section (validated against exact [100] bulk masses) with vertical field, bare (κ) and orbital Zeeman terms. Transport uses a dimension-generic RGF with Ando wave-matched normal leads and an $E = 0$ reflection invariant sign $\det r$ (kwant-equivalent; validated against closed-chain spectra to 10^{-13}). A test suite pins every exact claim. Δ is a *static* induced gap (the $\omega = 0$ limit of the standard tunneling self-energy); parent-gap field suppression enters via $\Delta_{ind} \leq \Delta_p(B)$ and metallization via the quasiparticle weight Z [7, 23, 24] (Supplement S6); the full dynamical self-energy is solved as a control in Sec. 6 and validates the static- Z treatment directionally while lowering its numbers by 24–30%. All caricature choices (GL/Zee-man-linear parent gaps, iid onsite disorder with correlation length dx , hard-wall LK geometry) are stated where used; complete per-figure parameter tables, seed families, and discretization/length/seed/grid convergence studies are in Supplement S4–S5.

3 Conduction band: three independent obstructions

3.1 Spin-orbit shortfall

Maximizing the topological gap over $(\mu, B \leq 2\text{ T})$ at measured intrinsic Rashba strengths gives $0.16\text{ }\mu\text{eV}$ (typical, $\alpha = 10^{-4}\text{ eV \AA}$) to $1.5\text{ }\mu\text{eV}$ (optimistic, 10^{-3}): below dilution-fridge protection scales and removed by onsite disorder far weaker than realistic interface fluctuations. Reaching a robust $20\text{ }\mu\text{eV}$ requires $\alpha \approx 0.015$ (idealized) to $0.021\text{ eV \AA}$ (with parent-gap suppression $\Delta(B)$)—an optimistic lower bound given unmodeled orbital and metallization corrections. Micromagnet arrays [8, 9] reach this scale in principle; a 2025 InAs experiment measured synthetic $\alpha = 0.022\text{ eV nm} = 0.22\text{ eV \AA}$ [10]—an order of magnitude *above* our silicon requirement, though demonstrated in a lighter-mass material with a mature hybrid stack, and with micromagnet fringe-field disorder unmodeled here. “Engineered-electron silicon” is therefore marginal rather than absurd—but the valley physics below stacks on top of it.

3.2 The $g = 2$ catch-22

Since $E_Z(2\text{ T}) = 116\text{ }\mu\text{eV}$ for $g = 2$, an induced gap $\Delta \geq 116\text{ }\mu\text{eV}$ cannot reach the topological phase for $B \leq 2\text{ T}$ within the single-channel $g = 2$ model; in the SOC-limited regime relevant to engineered devices, increasing interface transparency *reduces* the achievable gap, with an optimal parent gap $\Delta_0 \approx 98\text{ }\mu\text{eV}$ once $\Delta(B)$ suppression is included. This sharpens, for $g = 2$, the weak-coupling optimum identified by Cole, Das Sarma and Stanescu [7]. Pauli-limited parents satisfy the constraint automatically (Sec. 4.2).

3.3 Valley physics: an exact equivalence and step junctions

Equivalence theorem. With inter-valley pairing and *any* uniform VO phase, the wire maps unitarily onto two decoupled bands at $\mu \mp |\lambda|$ with pairing $\pm\Delta$ (proof in Supplement S2; the key step is the exact invariance of the inter-valley pairing under the valley rotation, $U_\phi D U_\phi^T = D$; test-suite tolerance $10^{-6}\text{ }\mu\text{eV}$ on every eigenvalue, spot checks $3 \times 10^{-10}\text{ }\mu\text{eV}$, for valley-scalar and phase-locked Dresselhaus SOC alike): smooth interfaces are benign, and valley splitting $\gtrsim 100\text{ }\mu\text{eV}$ buys single-valley topology, as the simple wedge picture suggests.

Step junctions. Real interfaces are stepped. That steps shift the VO phase by $2k_0\delta z$ is established quantum-dot physics [12, 11]; the new statement here is its consequence under inter-valley *pairing*: projecting the pairing onto the local valley bands gives each band an order parameter of phase $-\phi(x)$ (Supplement S3, exact), so each single-atomic step, $\Delta\varphi = 2k_0(a/4) \approx 0.85\pi$, is a Josephson junction accidentally close to π in silicon. The bound-state depth is determined numerically; the transparent short-junction formula $\Delta_t |\cos(\Delta\varphi/2)|$ tracks it to 10–25% (Supplement Table 1). A controlled comparison (same-sign staircase vs sign-randomized steps of equal magnitude) shows net phase *winding* is irrelevant at realistic densities; the damage is per-step junction physics: a single step binds a subgap state at $4.0\text{ }\mu\text{eV}$ (18% of the $22\text{ }\mu\text{eV}$ clean gap), deepening to a Kitaev domain-wall zero mode at $\Delta\varphi = \pi$ [Fig. 1(b)]. Ensembles at 50 nm step spacing retain only ~ 0.7 – $2.3\text{ }\mu\text{eV}$ (median, L -dependent upper bounds). Bunched double-height steps ($\Delta\varphi = 1.7\pi \equiv -0.3\pi$) are $\sim 2\times$ milder—step-bunched vicinal surfaces are preferable—while VO-amplitude suppression near steps [12] worsens the damage; wires laid along step edges avoid it entirely ($|\cos\chi|$ scaling). Smooth phase gradients are a real but subdominant second channel, suppressed at large valley splitting. Transport (Sec. 5) adds a warning: step-dense wires become Anderson insulators whose end-to-end transport gap *grows* with length ($0.5/5.6/23\text{ }\mu\text{eV}$ at $L = 1.5/3/6\text{ }\mu\text{m}$) while localized subgap states persist—a hard-looking transport gap does not certify protection. The falsifiable

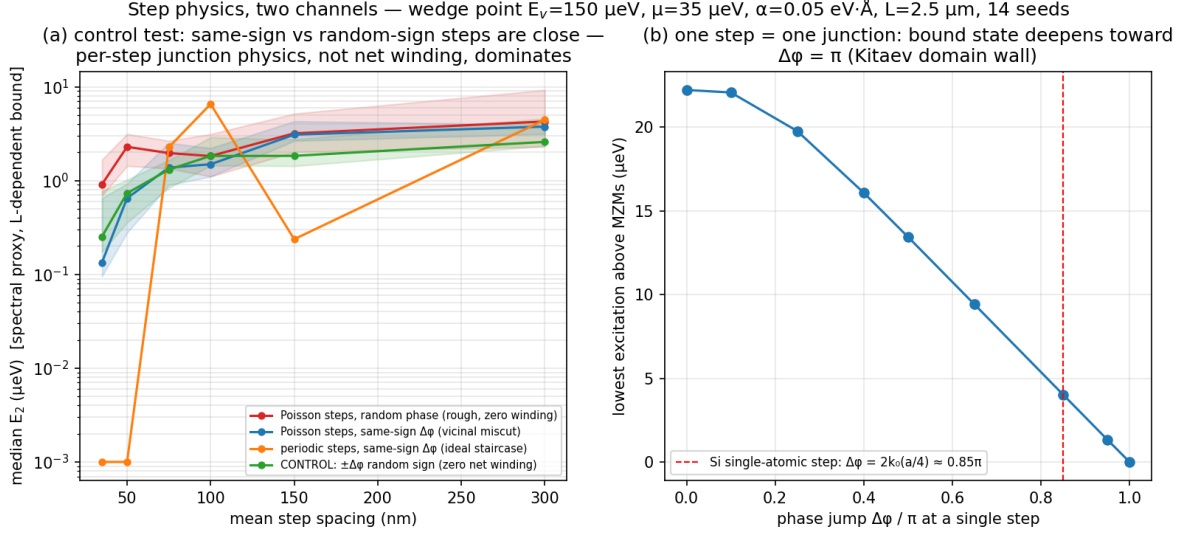


Figure 1: Step physics in the inter-valley-paired electron wire. (a) Controlled scenario comparison at fixed jump magnitude: same-sign (vicinal) and sign-randomized (zero-winding control) staircases are statistically indistinguishable—per-step junction physics, not winding, dominates. (b) A single VO phase step is a Josephson-like junction; the Si single-step value 0.85π sits near the Kitaev domain-wall point π .

consequence: induced-gap hardness in proximitized Si *electron* devices should degrade with miscut/step density from the very first step, testable by gap spectroscopy on standard miscut wafer series; holes, having no conduction-band valleys, are not subject to this particular mechanism (they remain subject to every other interface imperfection).

4 Valence band

4.1 Luttinger–Kohn constraints: the α – g covariation

Quasi-1D hole channels in Si FinFETs have measured strong, gate-tunable SOC [15, 17, 14] and anisotropic g -tensors [16]. Our four-band LK fin model reproduces the measured spin-orbit-length window ($l_{so} = 32\text{--}101 \text{ nm}$) and yields the constraint the free parameter box hides [Fig. 2]: the vertical field that builds α (up to $0.054 \text{ eV}\cdot\text{\AA}$) suppresses the wire-axis g_x from 1.1 to 0.5 while raising g_z to 2.4–4.2, with $m^* \approx 0.44$ and the SOC axis in-plane transverse. A seven-geometry bracket (7–16 nm cross sections, 3–30 MV/m) keeps $g_x \in [0.4, 1.1]$: the measured $g_{xx} \approx 1.9\text{--}2.3$ [16] is *not* reproducible in four-band hard-wall LK, so all platform numbers below are carried under both the LK and empirical tensors; resolving the discrepancy requires six-band $k \cdot p$ with device electrostatics (top remaining theory item).

4.2 Parents, orientation, and the surviving designs

Field-orientation maps [Fig. 3], using the tilted-field topological criterion of Osca, Ruiz and Serra [13], under both g -tensors and two parents give the design rules: (i) an **Al film** parent is locked to in-plane fields but strong— $34 \mu\text{eV}$ metallization-renormalized at the center point, up to $55 \mu\text{eV}$ bare over orientations (Fig. 3, not renormalized)—*if* a hard-gap Al/Si-hole interface can be fabricated

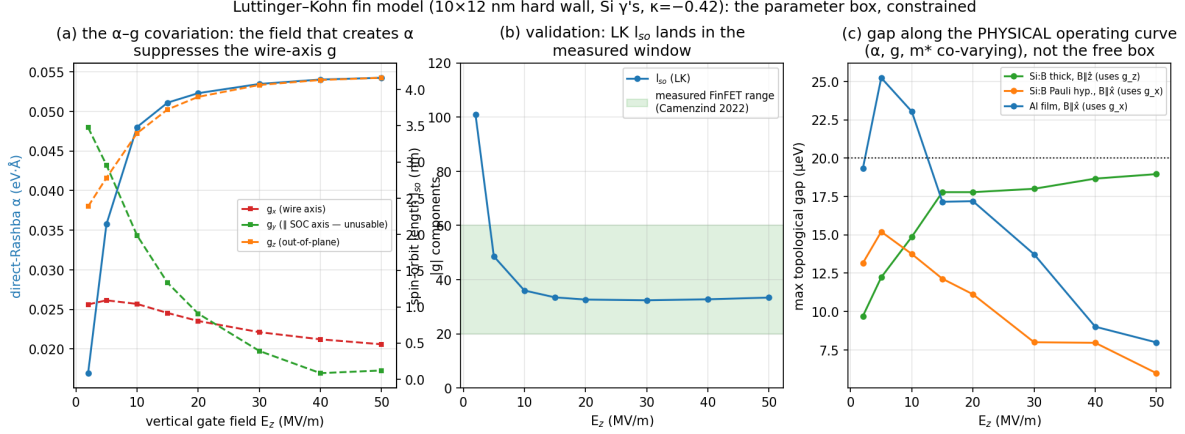


Figure 2: Luttinger–Kohn constraint on the hole platform: (a) α – g covariation under the gate field; (b) validation of l_{so} against FinFET measurements; (c) accessible gap along the physical operating curve for three parent/orientation scenarios.

(none exists); (ii) a **thick Si:B** parent [18, 19, 20, 21] is orientation-tolerant and, using only measured critical fields ($B_{c2} \approx 0.1$ – 0.4 T, orbital-limited), supports 10 – 19 μeV at fields tilted toward out-of-plane (riding g_z ; the center point is 11.0 μeV bare / 9.8 – 10.1 renormalized on finer optimizer grids, Supplement S5, moves over 8.0 – 10.6 μeV under $\pm 20\%$ parent-model variations, and falls to 7.6 μeV under the dynamic self-energy control of Sec. 6); the in-plane wire-axis direction is g -starved (3 – 9 μeV). (iii) The **thin-film Si:B** route turns the “Pauli-limited” hypothesis into a fabrication target: with coherence lengths from measured perpendicular critical fields ($\xi = 29$ – 57 nm), films of $d \lesssim 18$ – 36 nm have parallel orbital critical fields above the Pauli limit $B_P = 1.11$ T, and $d = 10$ – 20 nm films support **21**–**27** μeV (modulo unmodeled $T_c(d)$ suppression). Quasiparticle poisoning bounds the role of the measured-field stack: at operating fields the Si:B parent gap is 29 – 33 μeV , so $x_{qp} = 10^{-6}$ requires $T_{\text{eff}} \leq 25$ – 29 mK (Al reference: 130 mK)—a detection platform, not a parity-coherent qubit, absent parent-gap engineering.

5 Transport-level verification

RGF transmission with normal leads validates the clean wire (transport gap 37.3 vs bulk 36.8 μeV) and provides $\text{sign det } r$ as a topological invariant. Under onsite disorder the spectral proxy (third-smallest $|E|$) is *conservative* at moderate strength (localized states carry no current): the engineered-electron case’s half-gap threshold moves from $W \approx 410$ to ≈ 548 μeV , and breakdown is abrupt—a class-D zero-energy transmission anomaly coinciding with invariant sign flips—rather than a soft closing [Fig. 5]. The same machinery extended to eight-orbital valley cells yields the Anderson-insulator warning of Sec. 3.3.

6 Realism controls, master assumption table, and operational criteria

Review round 5 added four controls (Supplement S9; `realism.py`, `orbital.py`, `pairing_mix.py`, `morphology.py`). (i) *Dynamic self-energy*: replacing the static Δ with the full $\Sigma(\omega)$ and re-optimizing over tunnel coupling Γ gives center-point gaps of 7.6 μeV (measured Si:B, $\Gamma^* = 10$ μeV),

Field-orientation maps ($\mu=0$): star = in-plane along wire; triangle = out-of-plane.
 LK says: out-of-plane B (g_z) with a thick parent; the in-plane/wire direction is g-starved.

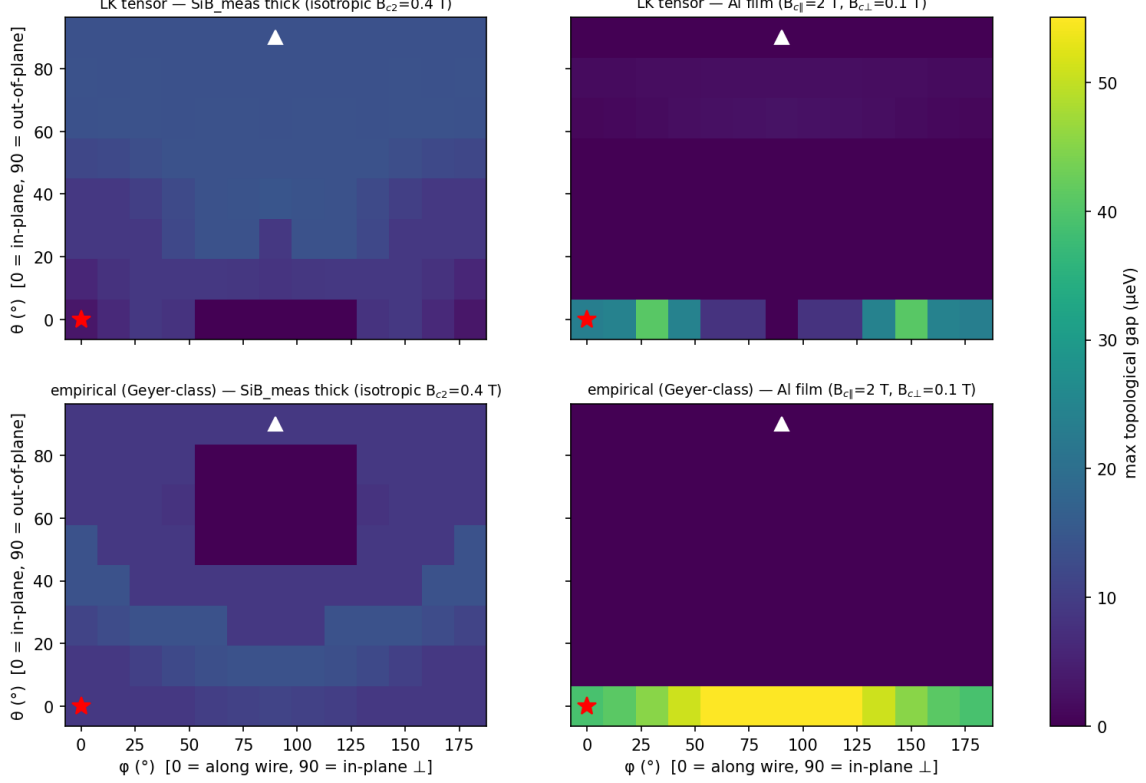


Figure 3: Maximum topological gap vs field orientation for LK (top) and empirical (bottom) hole g -tensors, with thick measured-Si:B (left) and Al-film (right) parents. Star: in-plane along the wire; triangle: out-of-plane.

14.9 μeV (Pauli Si:B, $\Gamma^* = 30$), and 24.2 μeV (Al, $\Gamma^* = 45$) — 1.4–2.1 $\times$ below the static bare values and 24–30% below the static-renormalized ones, which are hereby demoted to upper bounds; Dynes broadening adds a $\gamma_D/\Delta_p \lesssim 10^{-3}$ requirement for 50 mK operation. (ii) *Orbital coupling*: Peierls-phase controls show $< 0.05\%$ gap suppression at fin widths for the tilted Si:B design — the orientation maps survive. (iii) *Pairing-channel mixing*: intra-valley admixture η never weakens the step-junction mechanism (the step reappears as a pairing-phase junction; $2\times$ deeper bound state by $\eta^* \approx 0.24$) while healing staircase ensembles — our $\eta = 0$ ensembles are conservative, and $\eta \ll \eta^*$ is expected since intra-valley pairing requires $2k_0$ momentum transfer. (iv) *Morphology and correlated disorder*: with miscut-derived terrace ensembles, device *yield* (25–33% failure fraction at 0.05–0.1° miscut), not the median gap, is binding; wires along step edges or step-bunched templates empty the failure tail. Correlated $\mu(x)$ disorder at 50 μeV RMS collapses the hole-platform median gap to 3.5–6.8 μeV where the iid baseline retains 29 μeV : smooth electrostatics is the binding fabrication requirement, and our earlier iid thresholds overstate robustness by 4–8 $\times$ at fixed RMS.

6.1 Operational criteria

“Usable” is not a single threshold; we therefore evaluate every platform scenario of Table 1 against three named, quantitative tiers.

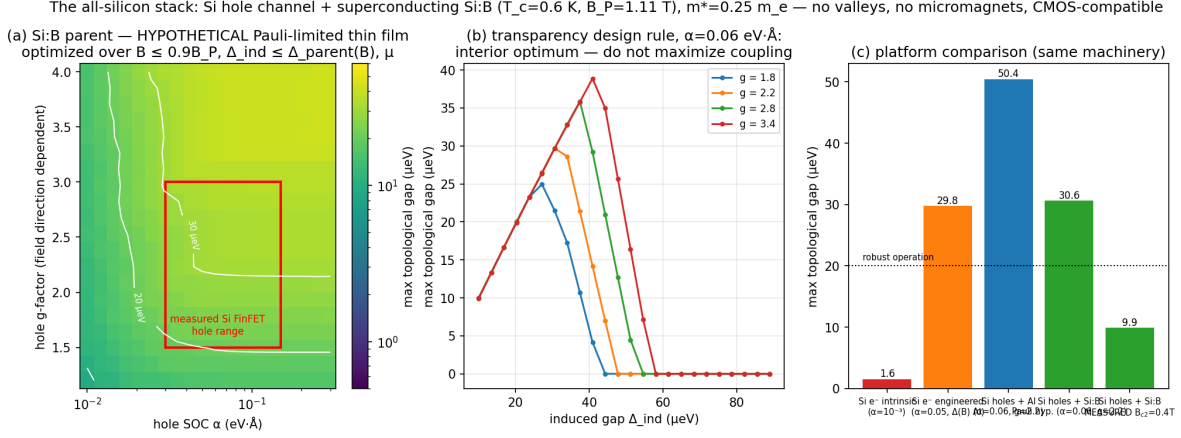


Figure 4: The all-silicon stack under the (hypothetical) Pauli-limited Si:B parent: (a) gap over the empirical (α, g) box; (b) the transparency design rule (interior optimum); (c) platform comparison including the measured-field Si:B scenario.

Table 1: Master assumption table. One row per platform scenario: parent superconductor and its demonstration status, g -tensor source, field orientation, operating field B^* , assumed critical field, parent gap at the operating field $\Delta_p(B^*)$, induced gap Δ_{ind} , spin-orbit strength α , g used, chemical potential μ , and the bare and metallization-renormalized topological gaps. Grid-converged values are used where they exist (Supplement S5). [†] = not experimentally demonstrated.

Platform	Parent (demonstrated?)	g source	B orientation	B^* (T)	assumed B_c (T)	$\Delta_p(B^*)$ (μ eV)	Δ_{ind} (μ eV)	α (eV·Å)	g used	μ (μ eV)	bare gap (μ eV)	renorm. gap (μ eV)
(1) Si e^- , intrinsic SOC	Al-class film [†] (partial: interface)	n.a. (scalar g)	along wire	≤ 2 (opt.)	—	— (static Δ)	90 (assumed)	10^{-3} (opt. intrinsic)	2.0	opt.	1.55	n.a.
(2) Si e^- , engineered SOC [†]	Al-class film [†] (partial: interface)	n.a. (scalar g)	along wire	≤ 2 (opt.)	GL $\Delta(B)$ [†]	— (GL)	$\Delta_p(B)$; 50 base	0.05 [†]	2.0	opt.	29.8 (42.6 w/o $\Delta(B)$)	n.a.
(3) Si holes + thick Si:B, LK tensor	Si:B, laser-doped (yes: measured)	LK (4-band)	tilted $\theta \approx 30^\circ$, $\phi_B = 90^\circ$	0.33–0.38	0.4 (measured, isotropic)	29	11	0.054 (LK)	g_s 2.4–4.2	0	14.6 (best tilt)	n.a. ^m
(4) Si holes + thick Si:B, empirical tensor	Si:B, laser-doped (yes: measured)	measured (Geyer-class)	tilted $\theta \approx 30^\circ$, $\phi_B = 30^\circ$	0.33–0.38	0.4 (measured, isotropic)	29	11	0.06 (empirical)	(2.1, 2.35, 2.7); 2.2 center	0	13.8 (best tilt) / 11.0 (center)	9.8–10.1 (center)
(5) Si holes + thin-film Si:B, Pauli [†]	Si:B film, Pauli-limited (no: orbital-limited to date)	measured (Geyer-class)	in-plane, $B \parallel \hat{z}$	1.0	1.11 [†] (Pauli, B_P)	33	33	0.06 (empirical)	2.2	0	30.6	19.7–20.2
(6) Si holes + Al film, orientation max [†]	Al film (partial: film yes; Si-hole interface no?)	measured (Geyer-class)	in-plane, $\phi_B = 90^\circ$	≤ 2 (opt.)	2 () / 0.1 (⊥)	150 (at 1 T)	opt.	0.06 (empirical)	(2.1, 2.35, 2.7)	opt.	55.0 (41–55 over tensors)	n.a. ^m
(7) Si holes + Al film, center [†]	Al film (partial: film yes; Si-hole interface no?)	measured (Geyer-class)	in-plane, along wire	1.0	2 () / 0.1 (⊥)	150	opt. (≤ 150)	0.06 (empirical)	2.2	opt.	50.4	34.4

[†]Not experimentally demonstrated (hypothetical parent regime, undemonstrated interface, or engineered SOC).

^mThe field-orientation maps (Fig. 3) are *not* metallization-renormalized; renormalized values are quoted only at the optimizer center points. “opt.” = optimized over internally, value at the optimum not recorded in the numerical ledger. $\Delta_p(B^*)$ values from `qp_poisoning.py` at the quoted operating points; renormalization $Z = 1 - \Delta_{ind}/\Delta_p$ (Supplement S6). Row 5 film-design targets: $d = 10$ – 20 nm films support 21–27 μ eV if Pauli-limited (fig8.film.design).

Tier S (spectroscopy/detection). Topological gap $\Delta_{top} \geq 2k_B T_{eff}$ at $T_{eff} = 50$ mK, i.e. $\Delta_{top} \geq 8.6 \mu$ eV ($k_B \times 50$ mK = 4.3μ eV), and Majorana splitting $E_0 \leq 0.1 \mu$ eV at $L \leq 3 \mu$ m. The gap threshold ensures the subgap feature is resolvable against thermal smearing in tunneling spectroscopy at a routinely achieved electron temperature; the splitting threshold keeps the zero-bias peak unsplit at spectroscopic resolution and is met by our clean finite-wire checks ($E_0 = 3 \times 10^{-4} \mu$ eV at 2μ m, repository Fig. 1; hole-wire check $E_0 \approx 0$ at 3μ m, fig8.hole_wire_finite_check).

Tier N (nonlocality/fusion). Additionally $\Delta_{top} \geq 20 \mu$ eV and disorder half-gap threshold $W_{1/2} \geq 200 \mu$ eV. The 20μ eV value is the protection scale used throughout this paper (e.g. the α requirement of Sec. 3.1, fig3.alpha_needed_for_20ueV), giving $\gtrsim 4 k_B T$ headroom for braiding/fusion protocols at 50 mK; the disorder threshold demands a finite margin against realistic potential fluctuations, against which the transport-level $W_{1/2} \approx 548 \mu$ eV of the engineered-electron case (Sec. 5) and the hole-wire gap persisting to $W \approx 300 \mu$ eV (fig8.hole_wire_finite_check) are measured.

Fig 12 — transport gap of the engineered-Si wire ($\alpha=0.05$ eV·Å, $\Delta=50$ μeV , $B=1.5$ T, $\mu=0$, $L=2$ μm , $dx=2.5$ nm; normal leads $\mu_{\text{lead}}=2000$ μeV)

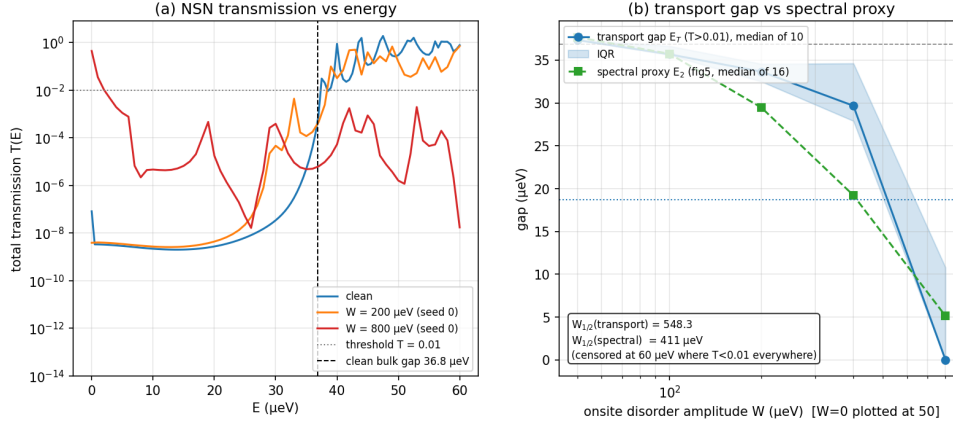


Figure 5: Transport vs spectral disorder thresholds (engineered-electron case): (a) transmission spectra; (b) median transport gap vs disorder with the spectral proxy overlaid.

Tier Q (parity-coherent qubit). Additionally a sustainable parent quasiparticle fraction $x_{qp} \leq 10^{-6}$ at $T_{\text{eff}} \geq 50$ mK given the parent gap *at the operating field*: 10^{-6} is the standard parity-lifetime requirement, and 50 mK is the coldest electron temperature routinely demonstrated for quasiparticle thermalization. From `qp_poisoning.py` (Supplement S7): the Si:B operating points ($\Delta_p(B^*) = 29\text{--}33$ μeV) require $T_{\text{eff}} \leq 25\text{--}29$ mK and therefore fail; the Al reference ($\Delta_p = 150$ μeV at 1 T) requires only $T_{\text{eff}} \leq 130$ mK and passes.

7 Discussion

Two measurements would do the most to settle the silicon question. First, the **parallel critical field of thin (≤ 20 nm) laser-doped Si:B films**: if it approaches the Pauli limit, as measured coherence lengths permit, the all-silicon hole stack supports $\gtrsim 20$ μeV gaps in a CMOS-compatible, same-crystal materials system; if films remain orbital-limited at ~ 0.4 T, the stack is capped near 10–15 μeV and detection-only. Second, the **wire-axis hole g -factor and SOC axis** in the actual gated geometry: our LK bracket and the published FinFET values differ by $2\times$ along the one direction that matters for film parents. Both measurements are standard characterization, neither requires a working topological device, and the electron-side miscut prediction offers a third, independent test of the framework. We caution that the step-junction mechanism, the orientation maps, and all platform numbers inherit the stated caricatures (single channel, hard-wall LK, parent-gap models, iid disorder, no orbital magnetic effects [25]); their domains of validity and the sensitivity of every headline number to them are quantified in Supplement S5–S6.

Data and code availability

All figures and quoted numbers regenerate from the committed code (`run_analysis.py`, `transport.py`, `transport_valley.py`, `lk_holes.py`, `qp_poisoning.py`, `convergence.py`, `tests.py`); the numerical ledger is `output/data/key_numbers.json`, and per-figure regeneration commands, the environment specification, and seed families are documented in `README.md` and Supplement S4/S10. The full repository — code, figures, numerical ledger, scan checkpoints, test suite, review-round corrections log, and this manuscript’s source — is public at

Table 2: Decision table: each platform scenario of Table 1 against the operational tiers. COND = passes the numerical thresholds only under the stated condition; [†] = scenario rests on an undemonstrated parent/interface (Table 1), so any PASS is contingent on its realization.

Platform (Table 1 row)	Tier S	Tier N	Tier Q	limiting condition
(1) Si e^- intrinsic	FAIL	FAIL	FAIL	gap $1.55 \mu\text{eV} \ll 8.6 \mu\text{eV}$
(2) Si e^- engineered [†]	COND	COND	COND	clean wire passes; interface steps reduce ensembles to 0.7–
(3) Si:B thick, LK tensor	PASS ^a	FAIL	FAIL	$T_{\text{eff}} \leq 25 \text{ mK}$ for $x_{qp} = 10^{-6}$; gap $< 20 \mu\text{eV}$
(4) Si:B thick, empirical tensor	PASS ^a	FAIL	FAIL	$T_{\text{eff}} \leq 25 \text{ mK}$ for $x_{qp} = 10^{-6}$; gap $< 20 \mu\text{eV}$
(5) Si:B thin film, Pauli [†]	PASS [†]	COND ^b	FAIL	parent hypothetical; renormalized gap 19.7–20.2 μeV strac
(6) Al film, orientation max [†]	PASS [†]	COND ^c	COND ^c	interface nonexistent; orientation-map values bare
(7) Al film, renorm. center [†]	PASS [†]	PASS ^{†,d}	PASS [†]	interface nonexistent

^aAt tilted field. The center-point renormalized value (9.8–10.1 μeV) clears 8.6 μeV with $< 2 \mu\text{eV}$ margin and dips to 7.6 μeV under $\pm 20\%$ parent-model variation (Supplement S5): the Tier-S pass is orientation- and parent-model-contingent. ^bBare value 30.6 μeV passes; the metallization-renormalized value 19.7–20.2 μeV sits at the 20 μeV line. ^cDisorder threshold not separately computed at Al-parameter operating points; gap criterion met on bare values only. ^d $W_{1/2}$ taken from the Si:B-class hole-wire check at comparable Δ_{ind} ; an Al-specific disorder run is future work.

<https://github.com/wtrevena/feasibility-of-creating-majorana-modes-in-silicon-nanowires>

An archival snapshot (tagged commit and DOI) will additionally be deposited on Zenodo at submission. [Zenodo DOI: to be inserted at submission.]

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References

- [1] A. Kitaev, Phys.-Usp. **44**, 131 (2001).
- [2] R. M. Lutchyn, J. D. Sau, S. Das Sarma, Phys. Rev. Lett. **105**, 077001 (2010).
- [3] Y. Oreg, G. Refael, F. von Oppen, Phys. Rev. Lett. **105**, 177002 (2010).
- [4] V. Mourik *et al.*, Science **336**, 1003 (2012).
- [5] H. Pan, S. Das Sarma, Phys. Rev. Research **2**, 013377 (2020).
- [6] M. Aghaee *et al.* (Microsoft Quantum), Phys. Rev. B **107**, 245423 (2023).
- [7] W. S. Cole, S. Das Sarma, T. D. Stanescu, Phys. Rev. B **92**, 174511 (2015).
- [8] M. Kjaergaard, K. Wölms, K. Flensberg, Phys. Rev. B **85**, 020503(R) (2012).

- [9] S. Turcotte *et al.*, Phys. Rev. B **102**, 125425 (2020).
- [10] arXiv:2505.06040 (2025).
- [11] M. P. Losert *et al.*, Phys. Rev. B **108**, 125405 (2023).
- [12] A. Hosseinkhani, G. Burkard, Phys. Rev. B **100**, 125309 (2019).
- [13] J. Osa, D. Ruiz, L. Serra, Phys. Rev. B **89**, 245405 (2014).
- [14] S. Bosco, B. Hetényi, D. Loss, PRX Quantum **2**, 010348 (2021).
- [15] L. C. Camenzind *et al.*, Nat. Electron. **5**, 178 (2022).
- [16] S. Geyer *et al.*, Nat. Phys. (2024).
- [17] B. Voisin *et al.*, Nano Lett. **16**, 88 (2016).
- [18] E. Bustarret *et al.*, Nature **444**, 465 (2006).
- [19] C. Marcenat *et al.*, Phys. Rev. B **81**, 020501(R) (2010).
- [20] J. E. Duvauchelle *et al.*, arXiv:1508.04075 (2015).
- [21] arXiv:2404.02748 (2024).
- [22] arXiv:2108.03031 (review).
- [23] F. Maier, J. Klinovaja, D. Loss, Phys. Rev. B **90**, 195421 (2014).
- [24] C. Reeg, D. Loss, J. Klinovaja, Phys. Rev. B **97**, 165425 (2018).
- [25] B. Nijholt, A. R. Akhmerov, Phys. Rev. B **93**, 235434 (2016).